



Department of Computer Science

SP.TOP: DATA SCIENCE AND ANALYTICS

COMP4381

Project Title:

Flight Delay Prediction Using Machine Learning Techniques

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Abstract:

Flight delays are a problem for the air transport industry. They affect how well things run make passengers unhappy and hurt the economy. This project looks at flight delays in five countries: the United States, the United Kingdom, New Zealand, Australia and Germany. We have a lot of data over 1,000 flight records and than 15 attributes. We want to use this data to predict when flights will be delayed. We will use a model that can tell us the probability of delays. This will help us make decisions and manage air traffic more efficiently. We also want to find out what causes flight delays.

Introduction:

The aviation sector is very important. It connects the world. Helps move people and goods around. This creates a lot of demand for air travel so managing flights is crucial. A delay is when a flight does not take off or land on time. The aviation industry faces challenges and flight delays are one of the biggest problems. They disrupt passenger schedules. Cost a lot of money. Other problems include weather, too much air traffic, plane malfunctions and poor scheduling. If we can predict delays airlines can make decisions that help both them and their passengers.

In years more people have been flying and new technologies have been developed. These technologies can help us predict flight delays. This project looks at the United States, the United Kingdom, New Zealand, Australia and Germany. We want to understand what causes delays in these countries. We have a lot of data including information about the time of year where the flightsre the weather. We will use this data to predict when flights will be delayed. We want to help airlines make decisions.

Background:

Flight delays are a problem in the aviation industry. A flight is considered late if it does not leave or arrive on time. When flights are delayed it causes trouble for passengers, such as missed connections. Delays also cost airlines a lot of money. Create problems at airports.

There are reasons why flights are delayed. These include weather, technical problems, busy airports and issues with airlines. Understanding these reasons is key to understanding delays. It can also help make air travel better for everyone.

Domain Explanation:

The aviation field is about managing flights between airports around the world. Each flight has a planned departure and arrival time. If the actual time differs from the planned time its considered a delay.

Flight information includes details like the airline, where the flight starts and ends flight time and how long the delay's. The weather and other outside factors can also affect how well a flight does.

Looking at flight data helps find patterns in flight delays. It also helps build computer models that can predict delays. This can improve decision-making in the aviation industry. Flight delays are an issue. Flight data is very important. Flight delays need to be analyzed.

Related Work (5 papers):

Study [1]:

This study checks how well six machine learning models work for predicting flight delays. The models are Decision Tree, Random Forest, SVC, Logistic Regression, KNN and Naive Bayes. To fix a problem with the data where one type of outcome happens often than the other we used techniques like Random Oversampling, SMOTE and ADASYN. We judged the models by how accurate they were their F1-score, MCC and ROC-AUC. We found that Random Forest with Random Oversampling and Decision Tree with SMOTE did the best with an accuracy of 90%. This study shows that balancing the data and using the ways to judge the models is important for making good predictions.

Air transport helps the global economy grow. It also helps with trade, tourism and job creation. With more flights happening every day there's a chance of flight delays. Delays cost airlines money. Make customers unhappy. So it's crucial to predict flight delays. This helps airlines make schedules and work efficiently. Machine learning has proven to be good at predicting delays. It helps airlines make choices. By using these models airlines can reduce delays. This makes passengers happier. Gives airlines an edge, over their competitors.

Study [2]:

Abstract :

This study addresses the problem of predicting flight delays exceeding 15 minutes using machine learning techniques, with the aim of reducing losses and improving flight management. The study relied on a real dataset from an airport in Turkey covering three years, including flight data such as departure time, number of passengers, and distance, as well as weather data such as temperature, atmospheric pressure, and humidity. Several algorithms were applied, including Logistic Regression, Naive Bayes, Random Forest, XGBoost, LightGBM, and CatBoost, in addition to neural networks. SMOTE was used to address data imbalances, and SHAP was used to analyze the significance of features. The results showed that the XGBoost model achieved the best performance with an accuracy of approximately 80%. This study can be linked to our project by using boosting models such as XGBoost and leveraging model interpretation techniques to understand the factors influencing delays.

Introduction :

Flight delays are among the most prominent challenges facing the aviation sector, significantly impacting airlines, passengers, and airports. These delays lead to substantial financial losses and decreased customer satisfaction. Therefore, it is crucial to develop models capable of predicting these delays in advance. This study aims to build a predictive model using machine learning techniques based on flight and weather data, in order to improve planning processes and reduce the impact of delays, thereby enhancing the efficiency of the air system and the passenger experience.

Study [3]:

Abstract :

This study aims to predict flight delays using both machine learning and deep learning, with the goal of improving prediction accuracy and addressing the complexity arising from multiple influencing factors. Large historical datasets from various sources, including flight, weather, and air traffic information, were utilized, along with data cleaning, missing value processing, and feature extraction. Algorithms such as Random Forest, Logistic Regression, Gradient Boosted Trees, and Support Vector Machines were employed, in addition to the Multilayer Perceptron (MLP) model as a deep learning tool. Hyperparameter tuning techniques were also applied to improve performance. The results showed that the MLP model achieved the highest accuracy of 98.83%, highlighting the superiority of deep learning in this area. This study can be used in our project by testing deep learning models and comparing them with machine learning models to select the most efficient model.

Introduction :

Flight delays are a complex problem that impacts operational efficiency in the aviation sector, causing financial losses and passenger dissatisfaction. Despite the availability of large amounts of historical data, predicting delays remains a challenge due to the multitude of influencing factors, such as weather, air traffic congestion, and operational issues. This study aims to use machine learning and deep learning techniques to analyze this data and discover patterns associated with delays. By developing accurate models, airlines can improve resource management, reduce delays, and enhance the passenger experience.

Study [4]:

Title of this study : Predicting Flight Delays Using Machine Learning.

This study is called Predicting Flight Delays Using Machine Learning. It talks about how flight delays are a big problem for the aviation industry. They cost a lot of money. This study uses data from US flights and weather data to predict when flights will be delayed. They used models, such as random forest and decision tree to make predictions. The results were very good. They found that the main causes of delays are: planes, problems with the airline problems with air traffic control and bad weather.

Citation:

Snell, K., Zurada, J., Hatami, Z., Olszak, C., & Kozak, J.

(2026). Predicting flight delays using machine learning.

In Proceedings of the 59th Hawaii International Conference
on System Sciences (pp. 275–284).

Study [5]:

This document asks a question: why do flights get delayed? It talks about predicting delays and measuring them. An on-time flight is one that takes off or lands within 15 minutes of the scheduled time. If a flight is delayed it can cause problems, including financial losses and customer dissatisfaction. This document talks about a model that can predict when flights will be delayed. They used data such as flight schedules, weather and runway congestion to make predictions. They found that the main causes of delays are: delays, bad weather and congestion on the runway.

Heres how each paper benefits my project:

These two papers are very helpful for my project. They help me choose the important factors select the best algorithm and understand the results. They also help me explain why one model is better than another. The United States, the United Kingdom, New Zealand, Australia and Germany are the countries we are looking at. Flight delays are a problem in these countries. We want to predict when flights will be delayed in these countries. We will use the data and models, from these papers to help us make decisions.

Citation:

Lemay, X., & Bastin, F. (2025). Prediction of airport on-time performance. arXiv:2601.00875.

Heres how each paper benefits my project:

These two documents are helpful in my project for selecting the most important factors, choosing the best algorithm, interpreting the results, and explaining why one model is better than another.

References:

1. Background: *Aviation System Performance Metrics* :<https://www.faa.gov>
2. Domain Explanation:*Airline On-Time Performance Data*:<https://www.transtats.bts.gov>
- 3.Study [1]: [Flight delay prediction: Evaluating machine learning algorithms for enhanced accuracy](#)
- 4.Study [2]: [Predictive Modeling of Flight Delays at an Airport Using Machine Learning Methods](#)
- 5.Study [3]: [\(PDF\) Airline Delay Prediction: A Comparative Study using Machine Learning and Deep Learning Models](#)
- 6.Study [4]: <https://hdl.handle.net/10125/111425>
- 7.Study [5]: <https://arxiv.org/abs/2601.00875>